

Module 4

Circuit Analysis in the S Domain

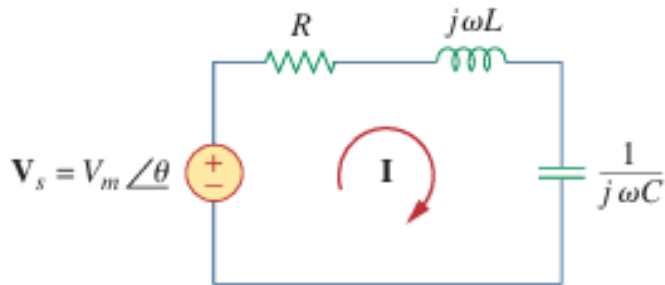
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Resonance circuit

Resonant circuits (series or parallel) are useful for constructing filters.

Consider the series RLC circuit



$$\mathbf{Z} = \mathbf{H}(\omega) = \frac{\mathbf{V}_s}{\mathbf{I}} = R + j\omega L + \frac{1}{j\omega C}$$

$$\mathbf{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

Resonance results when the imaginary part of the transfer function is zero.

$$\text{Im}(\mathbf{Z}) = \omega L - \frac{1}{\omega C} = 0$$

The value of ω that satisfies this condition is called the resonant frequency ω_0 . Thus, the resonance condition is

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

Resonance circuit

The frequency response of the circuit's current magnitude is as shown.

$$I = |\mathbf{I}| = \frac{V_m}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

$$P(\omega) = \frac{1}{2} I^2 R$$

The highest power dissipated occurs at resonance, when $I = V_m/R$, so that

$$P(\omega_0) = \frac{1}{2} \frac{V_m^2}{R}$$

At certain frequencies $\omega = \omega_1, \omega_2$ the dissipated power is half the maximum value; that is,

$$P(\omega_1) = P(\omega_2) = \frac{(V_m/\sqrt{2})^2}{2R} = \frac{V_m^2}{4R}$$

Hence, ω_1, ω_2 are called the half-power frequencies.

The half-power frequencies are obtained by setting Z equal to $\sqrt{2} R$ and writing

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2} R$$

Resonance circuit

Solving for ω ,

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$
$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

We can relate the half-power frequencies with the resonant frequency.

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

The width of the response curve depends on the bandwidth B , which is defined as the difference between the two half-power frequencies,

$$B = \omega_2 - \omega_1$$

Resonance circuit

The “sharpness” of the resonance in a resonant circuit is measured quantitatively by the quality factor Q .

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$$

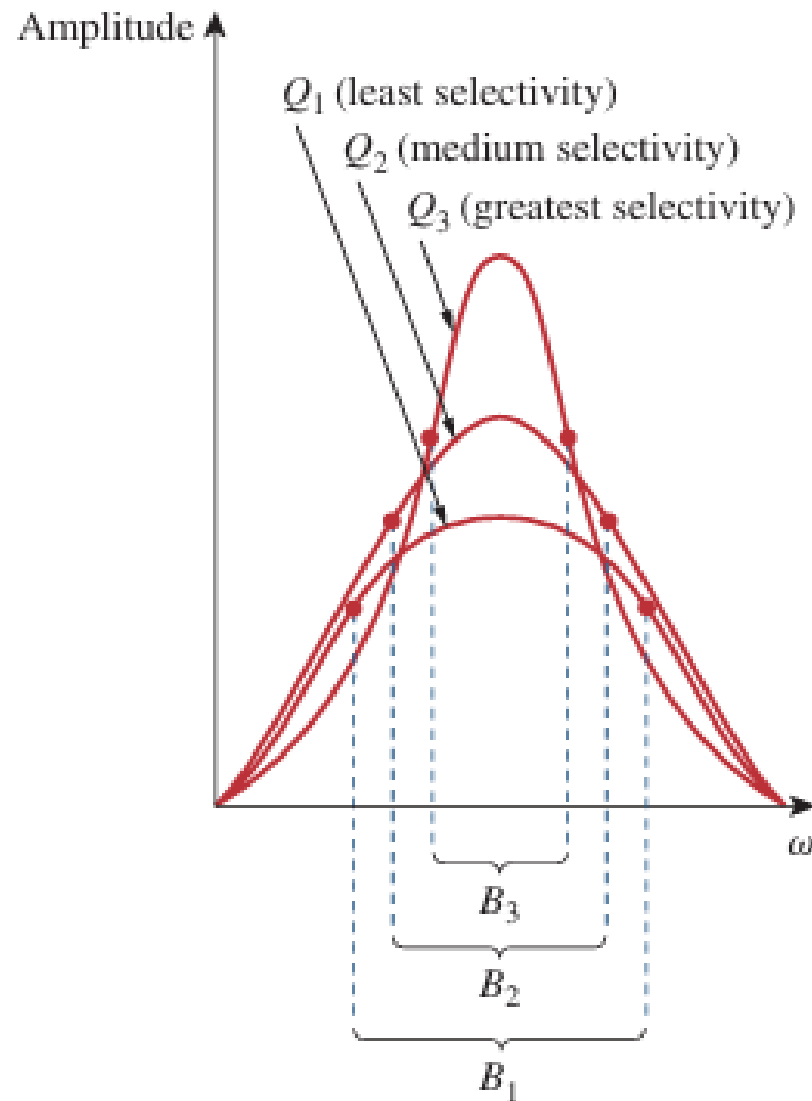
The quality factor is dimensionless. The higher the value of Q , the more selective the circuit is but the smaller the bandwidth.

$$B = \frac{R}{L} = \frac{\omega_0}{Q}$$

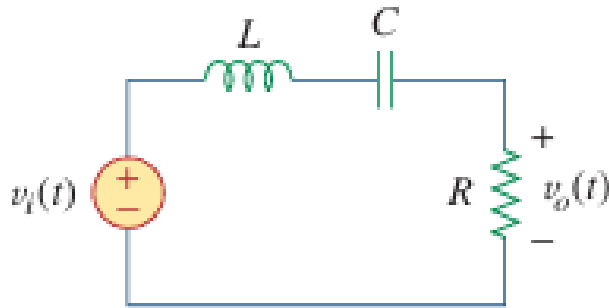
$$B = \omega_0^2 CR.$$

$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \quad \omega_2 \simeq \omega_0 + \frac{B}{2}$$

Resonance circuit



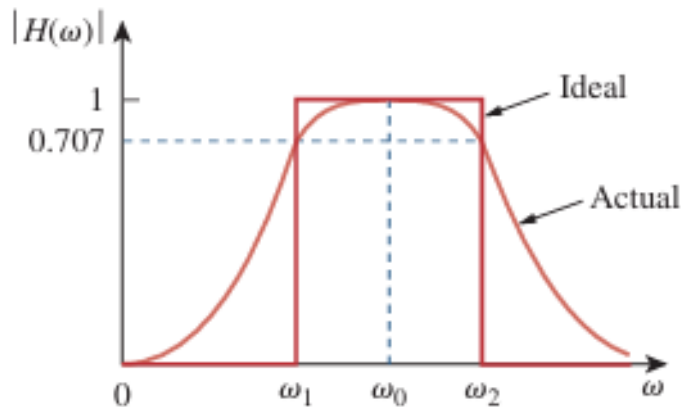
Bandpass Filters



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$$

Note that $H(0) = 0$ and $H(\infty) = 0$

Frequency response:



The bandpass filter passes a band of frequencies $\omega_1 < \omega < \omega_2$ centered on ω_0 the centre frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R}$$

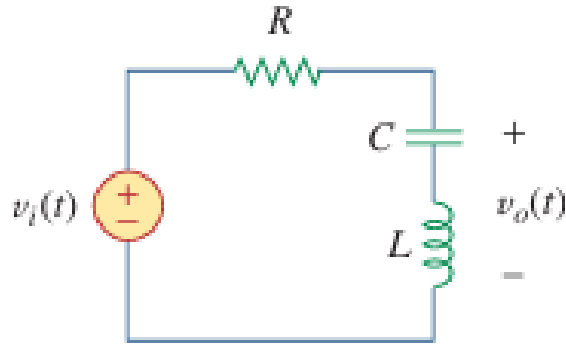
A bandpass filter can also be formed by cascading the lowpass filter (where $\omega_2 = \omega_c$) with highpass filter (where $\omega_1 = \omega_c$).

The bandpass filter is a series resonant circuit, the bandwidth, and the quality factor are related:

$$B = \frac{R}{L} = \frac{\omega_0}{Q}$$

$$B = \omega_0^2 C R.$$

Bandstop Filters

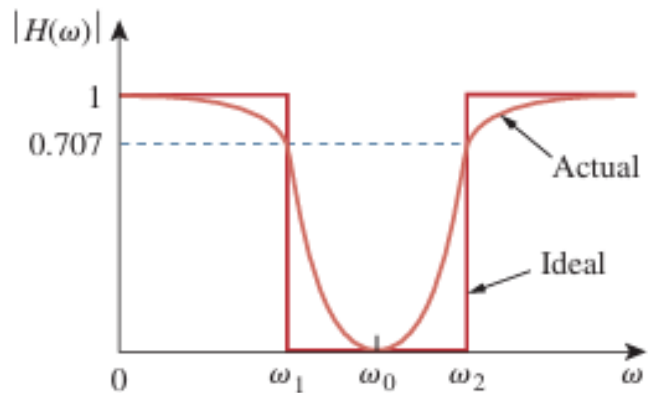


A filter that prevents a band of frequencies between two designated values (ω_1 and ω_2) from passing is known as a bandstop, band reject, or notch filter.

A bandstop filter is formed when the output RLC series resonant circuit is taken off the LC series combination.

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$$

Note that $H(0) = 1$ and $H(\infty) = 1$



$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Rejection frequency

$$\text{Bandwidth} = \omega_2 - \omega_1$$

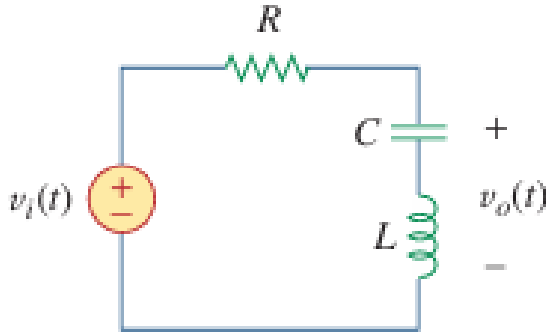
A bandstop filter is designed to stop or eliminate all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.

Passive Filters (Bandpass & Bandstop)

Resonant frequency, ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Quality factor, Q	$\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$	$\frac{R}{\omega_0 L}$ or $\omega_0 RC$
Bandwidth, B	$\frac{\omega_0}{Q}$	$\frac{\omega_0}{Q}$
Half-power frequencies, ω_1, ω_2	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$
For $Q \geq 10$, ω_1, ω_2	$\omega_0 \pm \frac{B}{2}$	$\omega_0 \pm \frac{B}{2}$

Passive Filters

1. If the bandstop filter in figure shown below is to reject a 200-Hz sinusoid while passing other frequencies, calculate the values of L and C. Take $R = 150 \Omega$ and the bandwidth as 100 Hz.



Express bandwidth in radians/sec, $B = 2\pi(100) = 200\pi \text{ rad/s}$

But,

$$B = \frac{R}{L} \Rightarrow L = \frac{R}{B} = \frac{150}{200\pi} = 0.2387 \text{ H}$$

Rejection of the 200-Hz sinusoid means that f_0 is 200 Hz.

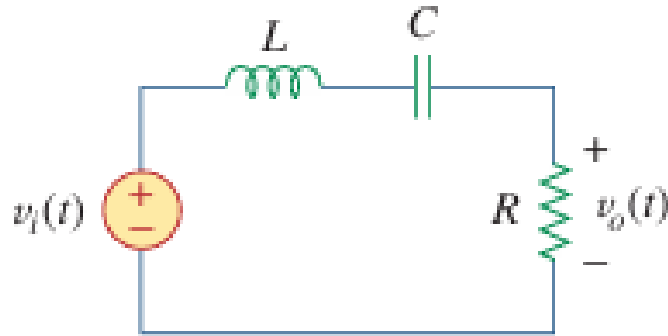
$$\omega_0 = 2\pi f_0 = 2\pi(200) = 400\pi$$

$$\omega_0 = 1/\sqrt{LC},$$

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(400\pi)^2 (0.2387)} = 2.653 \mu\text{F}$$

Passive Filters

2. Design a bandpass filter of the form shown in figure with a lower cutoff frequency of 20.1 kHz and an upper cutoff frequency of 20.3 kHz. Take $R = 20\text{k}\Omega$. Calculate L , C , and Q .



Answer:

15.915 H, 3.9 pF, 101.

Next class: Module 5